SQL Fundamentals: SELECT & FILTERING

① Selecting & Sorting

id	first_name	last_name	department

Table employees

id	first_name	last_name	department	salary	hire_date	city
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-5	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

01. Retrieve all columns from employees table

```
SELECT *
FROM employees;
```

→

id	First Name	Last Name	department	Salary	hire_date	City
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-5	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

Q2 Find all unique department

~~IT~~
~~HR~~
~~Finance~~
~~Marketing~~

Department
IT
HR
Finance
Marketing

~~SELECT DISTINC~~

```
SELECT DISTINCT(Department)
FROM employees;
```

Department
IT
HR
Finance
Marketing

Q3 Retrieve first and last names order by sala
descending

```
SELECT first_name,
       last_name
FROM employees
ORDER by salary desc;
```

first_name	last_name
Eva	Najdeo
Alice	Green
Cathy	Zely
David	Mokeng
Brain	Lee

04 Retrieve the top3 highest-paid employees.

~~SELECT *~~
~~FROM employees~~

```
SELECT id,
       first_name,
       last_name,
       salary
FROM employees
ORDER BY salary DESC
LIMIT 3;
```

id	First_name	last_name	salary
~~5~~ 5	~~Marie~~ Eva	Naidoo	70000
1	Alice	Green	70000
3	Cathy	Zulu	65000

FILTERING with WHERE

05 Find employees in IT department

```
SELECT id,
       first_name,
       last_name,
       department
FROM employees
WHERE department = 'IT';
```

id	First_name	last_name	~~department~~ ~~salary~~
1	Alice	Green	IT
5	Eva	Naidoo	IT

06 Find employee in Finance with Salary >60000

```
SELECT id,
        first_name,
        last_name,
        department,
        Salary
FROM employees
WHERE department IN 'Finance'
AND salary > 60000;
```

id	first_name	last_name	department	Salary
3	Cathy	Zulu	Finance	65000

07 Find employees in HR or Marketing

```
SELECT id,
        first_name,
        last_name,
        department
FROM employees
WHERE department IN ('HR', 'Marketing');
```

id	first_name	last_name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

08 Find employees not in IT

```
SELECT id,
        first_name,
        last_name,
        department
FROM employees
WHERE NOT department = 'IT';
```

id	first_name	last_name	department
2	Brain	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

09 Find employees in IT, HR or Finance

```
SELECT id,
        first_name,
        last_name,
        department
FROM employees
WHERE department IN ('IT','HR', 'Finance');
```

id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	Eva	Naidoo	IT

10 Find employees in IT with salary > 65000 and city Johannesburg

```
SELECT id,
       first_name,
       last_name,
       department,
       salary,
       city
FROM employees
WHERE ~~department~~ salary > 65000
AND city = 'Johannesburg';
```

id	first_name	last_name	department	salary	city
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Naidoo	IT	72000	Johannesburg